

IDEAL RAFUNA

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idealrafuna.com

Systems-focused Aerospace and Mechanical Engineer with experience in simulation, digital twin systems, UAV telemetry, robotics, and data-driven modelling. Strong background in systems integration, real-time engineering tools, Python-based analysis, and multidisciplinary physical/software systems.

EDUCATION

MASTER OF SCIENCE: AEROSPACE ENGINEERING 12/2024

University of Illinois Urbana-Champaign (UIUC), Urbana Champaign

- Focused on systems integration, simulation, and technology management.
- Co-authored research in spectral analysis of turbulence, with manuscript submitted to Physics of Fluids.

BACHELOR OF SCIENCE: MECHANICAL AND AEROSPACE ENGINEERING 05/2023

Clarkson University, NY

- Minor in Mathematics
- Inducted into Sigma Gamma Tau Aerospace Honor Society

EXPERIENCE

UNIVERSITY FOR BUSINESS AND TECHNOLOGY

Lecturer & Aerospace Laboratory Supervisor 10/2025 to Current

- Deliver **10 lectures and laboratory instruction hours per week** across undergraduate and graduate engineering courses, including Computer Science I/II, Electrical Science, Mechatronics, Mobile Robots, Autonomous Robotics, and CAD.
- Teach and support approximately **70 students** across Computer Science I and II, combining programming instruction, laboratory exercises, assessment, and academic support.
- Develop laboratory modules and supervise student engineering projects in embedded systems, robotics, UAV design, mechatronics, and control systems.
- Oversee aerospace laboratory activities supporting UAV systems integration, robotics experimentation, and multidisciplinary student research.

UBT LABS

TelloTwin - UAV Digital Twin System 09/2025 to Current

- Designed and integrated a **3-component UAV digital twin architecture** combining DJI Tello telemetry acquisition, a FastAPI backend, and a React/Three.js browser dashboard for real-time flight monitoring.
- Implemented real-time telemetry streaming at **10 Hz**, supporting a **17-field flight-state model** including battery, altitude, attitude, GPS, speed, signal, and flight-status data.
- Built **12 REST endpoints and 1 WebSocket telemetry stream** for drone commands, connection status, telemetry history, CSV logging, and real-time dashboard updates.
- Integrated real DJI Tello support with simulation fallback, mission planning, 3D flight-path visualization, emergency controls, historical playback, and telemetry export.

ALBABRIDGE TECH

Lead Developer / Part-Time Product Engineering Project 01/2025 to Current

- Built a cross-platform Albanian learning ecosystem spanning **iOS, Android, web, and backend API systems**, using React Native/Expo, TypeScript, Node.js/Express, PostgreSQL, Prisma, and Supabase.
- Developed **11+ interactive learning games** and mobile experiences, including quiz, word, rhythm, culture, pronunciation, and gamified practice features.
- Engineered a real-time multiplayer quiz system supporting **up to 200 players per room**, with host authentication, room codes, synchronized game state, scoring, leaderboards, timed questions, reconnect handling, and per-player response tracking.

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN

Research Assistant (Prof. Leonardo P. Chamorro) 08/2023 to 12/2025

- Developed frequency-domain correlation methods using nonuniform discrete Fourier transforms for analysis of irregularly sampled time-series data.
- Co-authored manuscript on spatial and temporal wind coherence analysis (submitted to Physics of Fluids)
- Performed spectral analysis of multi-year wind velocity datasets from multiple U.S. airports to study spatial and temporal coherence in atmospheric flow

CLARKSON UNIVERSITY

Research Assistant (Prof. Craig Merrett)

08/2022 to 05/2023

- Conducted structural and aerodynamic analyses of butterfly wings using ANSYS Fluent and Structural.
- Performed CFD and structural simulations to evaluate aeroelastic behavior and aerodynamic performance.
- Prepared presentations for weekly updates to Professor Craig Merrett.

UNIVERSITY OF ILLINOIS URBANA-CHAMPAIGN & CLARKSON UNIVERSITY

Teaching Assistant & Tutoring — UIUC & Clarkson University

08/2020 to 12/2024

Tutor (2020–2025)

- Supported instruction and tutoring in engineering mechanics, aerodynamics, mathematics, and thermodynamics courses.
- Assisted with grading, laboratory supervision, and student academic support.

CLARKSON UNIVERSITY

NASA RASC-AL Team Leader

08/2022 to 02/2023

- Led a multidisciplinary student engineering team collaborating with partner universities including Khalifa University and RMIT University.
- Managed the thermodynamics subsystem team, coordinating technical work, progress reports, and integration with other project groups.
- Facilitated collaboration across international teams to support project development and technical coordination

PUBLICATIONS

- Caraway, B., Bogucki, R., **Rafuna, I.**, Reyna, M., Cheng, S., Chamorro, L.P. (2025).
Frequency-domain correlation of nonuniform time series: Coherence across distant wind stations.
Physics of Fluids
Co-authored research analyzing multi-year wind velocity records using nonuniform discrete Fourier transform for spectral correlation and coherence assessment across U.S. airports.
- Rexhaj, Y., Rexhepi, R., Jetullahu, A., **Rafuna, I.** (2024).
Multispectral Pedestrian Detection in Low-Light Conditions: Infrared, Visible, and Fusion-Based Approaches for CCTV Applications.
UBT International Conference – Mechatronics, Systems Engineering, and Robotics (MSER Track)
DOI: [10.33107/ubt-ic.2025.290](https://doi.org/10.33107/ubt-ic.2025.290)
Developed dual-modality adaptive pedestrian detection framework using YOLOv8 and brightness-guided Weighted Boxes Fusion (WBF) for real-time smart-city surveillance.

TECHNICAL SKILLS

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|---|---|
| <ul style="list-style-type: none">• Programming Languages
Python, C++, TypeScript, JavaScript | <ul style="list-style-type: none">• AI & Computer Vision
YOLOv8, multispectral detection, computer vision pipelines |
| <ul style="list-style-type: none">• Robotics & Systems Engineering
ROS, UAV telemetry systems, digital twin architectures, IoT system design | <ul style="list-style-type: none">• Software & Data Systems
React, Node.js, FastAPI, PostgreSQL, Supabase, Three.js |
| <ul style="list-style-type: none">• Simulation & Engineering Tools
Simulink, SolidWorks, CATIA, ANSYS Fluent, STK, RoboDK | <ul style="list-style-type: none">• Other Tools
Unity |

HONORS AND LEADERSHIP

- Ansys Systems Tool Kit (STK) Level 2 Master Certification — Ansys
- Member, Sigma Gamma Tau, Aerospace Engineering Honor Society